

A mixed stationary state outside the Riccati family in arXiv:1006.3328

Candidate counterexample packet

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Status: candidate counterexample likely valid. This gives a full negative answer to the literal question in Remark 1, subject to human review of the source’s intended terminology.

1 Source problem

Gardas and Puchala study a qubit coupled to an environment, with total Hilbert space $\mathbb{C}^2 \otimes \mathcal{H}_E \simeq \mathcal{H}_E \oplus \mathcal{H}_E$ and block Hamiltonian

$$\mathbf{H} = \begin{bmatrix} H_+ & V \\ V^\dagger & H_- \end{bmatrix}.$$

For an assignment map Φ , the reduced evolution is

$$T_t(\rho) = \text{Tr}_E \left(e^{-it\mathbf{H}} \Phi(\rho) e^{it\mathbf{H}} \right).$$

The source explicitly permits the standard uncorrelated assignment $\Phi(\rho) = \rho \otimes \omega$ for a fixed environment state ω .

For a bounded solution X of

$$XVX + XH_+ - H_-X - V^\dagger = 0, \tag{1}$$

Definition 3 calls the partial traces of the rank-one projectors onto vectors of the graph

$$\Gamma_X = \left\{ \begin{pmatrix} \psi \\ X\psi \end{pmatrix} : \psi \in \mathcal{H}_E \right\}$$

and its orthogonal complement Riccati states. Immediately after Theorem 1, Remark 1 asks [1]:

“The question whether all stationary states are Riccati states or if it is possible that stationary states exist that are not Riccati states is still open.”

At this point, some remarks, regarding theorem given above, should be made.

Remark 1. *The question whether all stationary states are Riccati states or if it is possible that stationary states exist that are not Riccati states is still open.*

Definition/Theorem context and the complete question in Remark 1 of the original paper, page 5.

2 Proof intuition

The source's own Sylvester example ($V = 0$) already makes the obstruction visible. Separate the spectra of H_+ and H_- ; then the Sylvester equation has the unique solution $X = 0$, so its graph construction yields only the two pure computational-basis states. But stationarity is affine: under a fixed product assignment with an environment state commuting with the conditional environment Hamiltonians, every diagonal mixture of those two states is stationary. A nontrivial mixture is therefore stationary but is not itself one of the rank-one graph states in Definition 3.

The model below is not an uncoupled artifact. Its interaction is the nonzero, non-scalar dephasing term $\sigma_z \otimes K$; only the off-diagonal block V vanishes.

3 Counterexample

Theorem 1 (Stationary need not mean Riccati). *There is a two-level open quantum system, with a two-dimensional environment and a standard product assignment, having stationary states that are not Riccati states in the sense of Definition 3 of the source. In particular, Remark 1 has a negative answer.*

Proof. Let $\mathcal{H}_E = \mathbb{C}^2$, put

$$K = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{H} = 2\sigma_z \otimes \mathbf{1} + \sigma_z \otimes K.$$

Thus $\mathbf{H}$ has a genuine conditional interaction $\mathbf{H}_{\text{int}} = \sigma_z \otimes K$, and in the qubit block decomposition

$$V = 0, \quad H_+ = 2\mathbf{1} + K = \text{diag}(3, 1), \quad H_- = -2\mathbf{1} - K = \text{diag}(-3, -1).$$

All operators are bounded and self-adjoint, as required in the finite dimensional special case of the source.

Equation (1) reduces to

$$XH_+ - H_-X = 0.$$

Write $H_+ = \text{diag}(\lambda_1, \lambda_2)$ and $H_- = \text{diag}(\mu_1, \mu_2)$. Entrywise, this equation is

$$(\lambda_j - \mu_i)x_{ij} = 0 \quad (1 \leq i, j \leq 2).$$

The coefficient matrix is

$$\begin{pmatrix} 3 - (-3) & 1 - (-3) \\ 3 - (-1) & 1 - (-1) \end{pmatrix} = \begin{pmatrix} 6 & 4 \\ 4 & 2 \end{pmatrix},$$

so every entry of X is zero. Hence $X = 0$ is the unique bounded solution of the Riccati equation.

For this solution, graph vectors and orthogonal-graph vectors are respectively $\begin{pmatrix} \psi \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ \phi \end{pmatrix}$. After normalization, their partial traces are independent of the nonzero choices of ψ and ϕ and equal to

$$P_0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_1 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Because no other Riccati solution exists, these are all the Riccati states for this Hamiltonian in the source's sense.

Now fix the product assignment

$$\Phi(\rho) = \rho \otimes \omega, \quad \omega = \frac{1}{2}\mathbf{1},$$

which is one of the assignments explicitly admitted by the source. For any $0 < a < 1$, let

$$\rho_a = \begin{pmatrix} a & 0 \\ 0 & 1-a \end{pmatrix}.$$

Both ρ_a and ω commute with the two tensor factors occurring in $\mathbf{H}$, and therefore

$$[\mathbf{H}, \rho_a \otimes \omega] = 0.$$

Consequently

$$e^{-it\mathbf{H}}(\rho_a \otimes \omega)e^{it\mathbf{H}} = \rho_a \otimes \omega \quad \text{and} \quad T_t(\rho_a) = \text{Tr}_E(\rho_a \otimes \omega) = \rho_a$$

for every t . Thus ρ_a is stationary. Since $0 < a < 1$, it is mixed and is neither P_0 nor P_1 , so it is not a Riccati state. This proves the claim. $\square$

4 Verification report

The argument is finite-dimensional and rests on four exact checks.

1. The model has the source's block form, with bounded $V = 0$ and self-adjoint $H_{\pm}$; its interaction $\sigma_z \otimes K$ is nonzero and non-scalar on the environment.
2. The four Sylvester coefficients are 6, 4, 4, 2, hence $X = 0$ is the unique solution.
3. Definition 3 applied to $X = 0$ gives exactly the two normalized states P_0, P_1 .
4. The displayed commutator vanishes, proving stationarity under one fixed, standard assignment map for every t .

The accompanying standard-library Python script verifies the coefficient calculation, the 4×4 commutator, and the partial trace exactly using rational arithmetic. It was run once with $a = 1/3$ and all assertions passed. The computation is only a check; the entrywise and commutator arguments above are the proof.

5 Terminology and limitations

This packet reads *Riccati state* literally as Definition 3 does in the context of Theorem 1: a partial trace of a rank-one projector from the graph or orthogonal graph associated with a bounded solution of the Hamiltonian's Riccati equation. Allowing an arbitrary operator X unrelated to the Riccati equation would trivialize the link to the Hamiltonian and is not the question posed after Theorem 1.

The mixed state ρ_a lies in the convex hull of P_0 and P_1 . If one redefines "Riccati states" to mean the convex hull of the states in Definition 3, then this particular example is inside that enlarged class. The source neither makes that definition nor states a convexified question. Accordingly, the counterexample is a complete negative answer to the literal open question, while the stronger convex-hull variant is not addressed here.

6 Bounded novelty check

The search through 11 August 2026 covered the run’s registry, solution, attempt, and proof-gap indexes; the exact sentence in Remark 1; the paper title and arXiv identifier 1006.3328 combined with *Riccati*, *stationary*, *counterexample*, and *open quantum*; and close stationary/Riccati phrase variants. It found the original paper and mirrors/versions of it, but no later source explicitly answering Remark 1. The source’s own $V = 0$ example lists only the two pure projections and does not mention their stationary mixtures. Novelty confidence is moderate pending a specialist search of non-indexed quantum-information literature; validity of the counterexample is independent of novelty.

7 Conclusion and review recommendation

The literal question in Remark 1 has a negative answer: even a finite-dimensional dephasing model with a unique Riccati solution has mixed stationary states outside the rank-one Riccati family of Definition 3. Promotion as a likely valid full counterexample is recommended, with human review focused on whether the intended term was silently meant to include a convex hull.

References

- [1] B. Gardas and Z. Puchala, *Stationary states of two-level open quantum systems*, arXiv:1006.3328. <https://arxiv.org/abs/1006.3328>.